Federated Graph-Based Multi-Agent Model Predictive Control for Low-Carbon Urban Traffic Signal Coordination

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Abstract—Urban traffic congestion is a critical challenge in modern metropolitan areas, imposing severe economic and environmental burdens. As cities expand, traditional traffic signal control systems, which predominantly optimise for vehicle throughput and delay, are increasingly inadequate as they neglect the escalating impact of vehicular carbon emissions. Intelligent Transportation Systems (ITS) must evolve to embed sustainability criteria directly into the real-time control loop. However, existing solutions encounter significant limitations. Centralised control architectures suffer from scalability bottlenecks and communication latency in large-scale networks, while distributed approaches often struggle to align local actions with network-wide goals. Furthermore, the transmission of real-time state data in cloud-based models raises substantial privacy and infrastructure concerns. To address these critical challenges, this paper presents the design, implementation, and evaluation of a novel federated graph-based multi-agent Model Predictive Control (MPC) system for sustainable urban traffic signal coordination. Our proposed architecture models the road network as a directed graph where intersections act as autonomous, privacy-preserving agents. Each agent solves a local convex quadratic programme over a finite predictive horizon, utilising a Long Short-Term Memory (LSTM) network for online multi-step demand forecasting. To align global sustainability objectives without sharing sensitive queue data, a Federated Averaging (FedAvg) mechanism is employed to periodically synchronise an adaptive carbon-penalty weight across all agents. Extensive evaluations driven by real-world datasets demonstrate the system's efficacy. Simulation results establish that the proposed carbon-aware federated MPC reduces total cumulative network emissions by 10.0% to 16.2% relative to a standard throughput-maximising baseline. This environmental benefit is achieved while maintaining queue stability and bounding the mean queue length to under 1.4 vehicles per agent, even when subjected to acute, 300% demand shocks. The system is computationally efficient, exhibiting $\mathcal{O}(H^3)$ scaling that executes well within a 0.2-second window for real-time deployment.

Index Terms—Federated Learning, Model Predictive Control, Traffic Signal Control, Intelligent Transportation Systems, Carbon-Aware Optimisation, Multi-Agent Systems.

I. Introduction

URBANIZATION and the rapid proliferation of vehicles have established traffic congestion as a paramount challenge in modern smart cities, manifesting in profound economic losses and environmental degradation [1], [2]. Within Intelligent Transportation Systems (ITS), the subdomain of adaptive traffic signal control has emerged as a critical lever for alleviating congestion by dynamically adjusting green splits in response to real-time traffic demand [3]. Historically, controllers prioritised macroscopic throughput and delay minimisation, fundamentally neglecting the escalating carbon footprint of vehicular emissions. As global mandates tighten around urban decarbonisation, the paradigm of traffic control must transition from purely delay-centric latency reduction to carbon-aware, multi-objective optimisation [4]. Consequently, embedding sustainability directly into the real-time control loop has become an urgent priority for next-generation ITS infrastructures [5], [6].

The integration of environmental objectives into real-time traffic management introduces a complex, often adversarial, optimisation problem. Throughput maximisation inherently encourages aggressive vehicle clearance, which generates substantial acceleration and traversal emissions, while restrictive control policies aimed at limiting flow paradoxically induce extended idling times and sprawling queue formations [7]. The impact of this unresolved trade-off is profound; unchecked vehicular idling contributes disproportionately to urban greenhouse gas inventories and localised degradation of air quality [8]. Furthermore, as traffic networks grow in scale and complexity, the non-linear dynamics of vehicle platoons under varying demand profiles render traditional, pre-timed, and isolated actuated controllers fundamentally inadequate [9]. The inability to systematically balance the macroscopic flow efficiency against granular, intersection-level carbon emissions threatens the sustainability objectives of modern smart city initiatives [10].

Existing advanced control systems encounter significant structural and operational limitations when attempting to resolve these concurrent objectives. Model Predictive Control (MPC) offers a theoretically robust framework for anticipating demand; however, classical centralised MPC architectures suffer from debilitating scalability bottlenecks in large-scale road networks and demand high-bandwidth, low-latency data streams [11]. Conversely, fully decentralised approaches frequently fail to achieve network-wide optimality due to myopic local actions. Data-driven methods, particularly those leveraging centralised Deep Reinforcement Learning (DRL), necessitate the continuous aggregation of raw queue counts and vehicle trajectories, raising severe privacy concerns regarding civilian mobility patterns and infrastructure

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vulnerability [12], [13]. Additionally, many existing multi-agent reinforcement learning formulations struggle with non-stationary environments and require exhaustive re-training when deployed across heterogeneous intersection topologies [14].

To circumvent these limitations, the synergy of graph-theoretic modelling, convex optimisation, and federated learning presents a formidable algorithmic solution. Graph neural structures naturally encode the spatial topology of road networks, facilitating the structured propagation of traffic states among interconnected intersections [15], [16]. When coupled with local convex MPC solvers, this allows for the rapid resolution of control sequences over finite predictive horizons using locally constrained quadratic programming (QP). Crucially, Federated Learning (FL) provides a robust mechanism for collaborative model consensus without the centralised pooling of sensitive raw data [17]. By distributing the computational burden to the edge (i.e., the signal controllers) and employing federated averaging (FedAvg), agents can collectively align on a shared parameter—such as an adaptive carbon-penalty weight—thereby achieving macroscopic coordination while preserving strict operational privacy and minimising communication overhead [18].

In this paper, we propose a novel federated graph-based multi-agent MPC architecture for sustainable urban traffic signal coordination. The road network is abstracted as a directed graph where each intersection agent independently solves a dual-source (throughput and idle) carbon-penalised convex QP to optimise its local green-time allocation. To enhance the predictive capability of the MPC, each agent is equipped with a local Long Short-Term Memory (LSTM) forecaster that predicts multi-step incoming demand online. Rather than transmitting raw traffic volumes, the agents participate in a periodic FedAvg protocol to synchronise a dynamic carbon-penalty coefficient. This coefficient is adaptively scaled based on local carbon budget violations, dynamically shifting the network's priority between congestion mitigation and emission reduction over time.

The specific, unique contributions of this study are summarised as follows:

- Dual-Source Carbon-Aware Convex MPC: Formulation of an intersection-level convex quadratic programme that explicitly balances fluid-flow queue dynamics with a dual-source (idle and throughput) vehicular emission model without sacrificing real-time solvability.
- Privacy-Preserving Federated Coordination: Introduction of a FedAvg-based parameter synchronisation protocol that aligns global network sustainability behaviour via an adaptive carbon-penalty coefficient, entirely eliminating the need for sharing raw real-time traffic volumes.
- Integrated Online LSTM Demand Forecasting: Deployment of robust, edge-executable LSTM forecasters at each intersection, continuously trained online, to provide high-fidelity multi-step demand predictions directly to the finite-horizon MPC solver.
- Empirical Validation of Pareto-Optimality: Rigorous demonstration of the proposed system's efficacy against severe operational shocks using real-world demand data, achieving up to 16.2% cumulative emission reduction while empirically tracing the Pareto frontier of the emission-queue trade-off.

The remainder of this paper is organised as follows. Section II comprehensively reviews related literature and identifies the critical research gaps. Section ?? details the mathematical formulation of the unified graph-theoretic, federated, and predictive control architecture. Section ?? presents the experimental setup, simulation results, and comparative discussions using real-world dataset profiles. Finally, Section V concludes the study and outlines potential avenues for future research.

II. Related Works

The intersection of adaptive traffic signal control and environmental sustainability has witnessed a proliferation of research in recent years, predominantly driven by advancements in deep learning, distributed optimisation, and federated systems. This section reviews 15 recent contributions published in IEEE Transactions between 2024 and 2026, categorizing them into three primary domains: carbon-aware traffic optimisation, predictive and graph-based control, and federated multi-agent learning systems.

A. Carbon-Aware and Multi-Objective Traffic Optimisation

Traditional signal control strategies have overwhelmingly focused on throughput and delay minimisation. However, recent literature emphasizes the critical need to incorporate environmental metrics. For instance, Patel et al. [19] proposed a framework integrating emissions modelling into adaptive control, demonstrating that isolated intersections can reduce traversal emissions by 8% through modified green splits. A complementary study by Kim et al. [20] introduced a dual-objective reinforcement learning agent that balances queuing delays with a penalty for acceleration-induced carbon outputs, though their approach was limited to pre-timed corridors. Similarly, Wang et al. [21] designed a heuristic signal coordination scheme that leverages vehicle-to-infrastructure (V2I) communication to smooth vehicle trajectories, indirectly lowering idling emissions [22]. While these studies successfully validate the feasibility of carbon-centric penalty functions, they frequently rely on centralised state aggregation, which scales poorly across expansive urban grids [23]. Expanding on this, Chen et al. [24] introduced an eco-driving integration protocol for signalized intersections, optimizing both signal phases and vehicle approach speeds, but acknowledged computational bottlenecks in real-time multi-intersection scenarios.

B. Predictive and Graph-Theoretic Control Architectures

To address the non-stationarity of urban traffic, predictive and structurally aware models have gained significant traction. Model Predictive Control (MPC) has been extensively explored, with Mohebifard and Hajbabaie [6] proposing a single-intersection MPC that optimizes signal timing in the presence of automated vehicles. Their findings confirmed superior queue management compared to actuated control, establishing a strong baseline for MPC efficacy. Extending predictive horizons, several researchers have integrated graph neural networks (GNNs) into the traffic forecasting loop. Liu et al. [16] developed a spatio-temporal contrastive learning-based adaptive graph augmentation method for traffic flow prediction, yielding highly accurate multi-step forecasts. Applying similar graph-theoretic principles, Zhang et al. [25] formulated a distributed MPC relying on a graph Laplacian matrix to share state predictions among adjacent nodes, reducing network-wide delay by 12% under heavy demand [26]. Furthermore, recurrent architectures, specifically Long Short-Term Memory (LSTM) networks, have been paired with graph structures to handle severe demand shocks, as demonstrated by the robust forecasting mechanism introduced by Singh et al. [27]. Despite these predictive enhancements, a majority of these graph-based MPCs remain purely delay-centric, failing to account for the carbon implications of their aggressive throughput optimisation [28].

C. Federated and Multi-Agent Reinforcement Learning

The inherent privacy and scalability challenges of centralized traffic management have accelerated the adoption of federated and decentralized multi-agent learning. Guo et al. [13] introduced a clustered federated learning architecture with adaptive pruning for 6G edge-intelligent transportation, illustrating that parameter sharing can achieve near-centralised performance while drastically reducing communication overhead. Parallelly, Liu et al. [15] proposed a position-aware structural knowledge sharing-based federated graph learning framework that enables intersection agents to coordinate effectively without exposing raw, localized traffic volumes. In the domain of deep reinforcement learning (DRL), hierarchical federated structures have been deployed successfully by Ma et al. [29] to train decentralized signal controllers across diverse urban topographies. A related study by Yang et al. [30] utilized FedAvg to synchronize the value networks of distributed DRL agents, enhancing convergence stability during periods of acute demand variation. Additional work by Li et al. [31] showcased a federated proximal policy optimization method for cross-domain signal control, significantly accelerating model generalization across differing intersection geometries.

D. Identification of Research Gap

While the reviewed literature demonstrates profound advancements in their respective silos—namely carbon-aware penalties, graph-based predictive control, and federated learning—a critical research gap persists at their intersection. Specifically, existing carbon-aware models primarily depend on centralized or isolated heuristic architectures, rendering them either unscalable or sub-optimal in dense networks. Conversely, state-of-the-art federated and graph-based multi-agent systems [13], [15], [16] lack explicit environmental objective functions, indiscriminately prioritizing throughput at the expense of idling emissions. The proposed system directly addresses this convergence gap by embedding a mathematically rigorous, dual-source carbon emission model entirely into a distributed, graph-coordinated convex MPC framework. By utilizing a completely novel federated synchronization of the adaptive carbon-penalty coefficient (γ), our architecture uniquely achieves macroscopic sustainability goals and robust queue stability without the necessity of sharing raw vehicular data, fundamentally distinguishing it from prior throughput-focused and centralized-carbon paradigms.

III. Proposed System

A. Overview

The proposed system integrates five technical pillars: (i) graph-theoretic network modelling, (ii) discrete-time fluid-flow queue dynamics, (iii) dual-source carbon emission modelling, (iv) convex MPC optimisation, and (v) federated learning coordination. The road network is represented as a directed graph; each node (intersection) operates as an autonomous agent that solves a local convex programme at every control step. The objective function simultaneously penalises queue lengths and estimated carbon emissions. Online LSTM forecasters improve the demand forecast supplied to each MPC solver. Periodically, a lightweight federated server synchronises the emission-penalty coefficient γ across all agents, aligning global sustainability behaviour without transmitting queue states or vehicle counts.

B. Graph-Theoretic Network Modelling

The urban traffic network is modelled as a directed graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}) \tag{1}$$

where $\mathcal{V}$ is the set of N intersection nodes and $\mathcal{E}$ is the set of directed road edges connecting adjacent intersections.

The node set is indexed as:

$$\mathcal{V} = \{1, 2, \ldots, N\} \tag{2}$$

where N denotes the total number of signalised intersections in the simulated network.

Topological connectivity is encoded in an adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$:

$$A_{ij} = \begin{cases} 1, & \text{if } (i, j) \in \mathcal{E} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where A_{ij} equals one if a directed road links intersection i to intersection j, and zero otherwise.

The graph Laplacian $\mathbf{L} \in \mathbb{R}^{N \times N}$ is:

$$\mathbf{L} = \mathbf{D} - \mathbf{A} \tag{4}$$

where $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_N)$ is the diagonal degree matrix with $d_i = \sum_j A_{ij}$, and $\mathbf{A}$ is the adjacency matrix from (3). The Laplacian captures the network's connectivity structure and underpins spectral analysis of distributed consensus convergence.

The spectral gap $\lambda_2(\mathbf{L})$ (second-smallest eigenvalue of $\mathbf{L}$) governs the speed of consensus among agents:

$$\lambda_2(\mathbf{L}) > 0 \iff \mathcal{G} \text{ is connected} \tag{5}$$

where $\lambda_2(\mathbf{L})$ equals zero for disconnected graphs and grows with network density, directly bounding the FedAvg convergence rate.

C. Traffic Flow and Queue Dynamics

For each agent $i \in \mathcal{V}$, the queue evolves under a discrete-time fluid-flow model:

$$q_{i,t+1} = \max(0,\ q_{i,t} + d_{i,t} - \mu_i\, g_{i,t}) \tag{6}$$

where $q_{i,t} \geq 0$ is the vehicle queue at intersection i at time t; $d_{i,t}$ is the incoming demand (vehicles arriving); μ_i is the saturation capacity (vehicles served per full-green step); and $g_{i,t} \in [0, g_{\max}]$ is the allocated green-time fraction.

To preserve convexity of the MPC formulation, the $\max(\cdot)$ operator is relaxed into dual inequality constraints:

$$q_{i,t+1} \geq q_{i,t} + d_{i,t} - \mu_i\, g_{i,t} \tag{7}$$

$$q_{i,t+1} \geq 0 \tag{8}$$

where (7) enforces conservation of unserved vehicles and (8) ensures non-negativity of the queue state. Because the objective strictly minimises q, these inequalities bind tightly at the optimum, recovering (6).

The effective service rate $s_{i,t}$ (vehicles cleared per step) is:

$$s_{i,t} = \mu_i\, g_{i,t} \tag{9}$$

where μ_i is the per-agent capacity and $g_{i,t}$ is the green-time fraction determined by the MPC solver.

D. Carbon Emission Modelling

The total emission $E_{i,t}$ at intersection i during timestep t comprises two sources:

$$E_{i,t} = E_{i,t}^{\text{idle}} + E_{i,t}^{\text{throughput}} \tag{10}$$

where $E_{i,t}^{\text{idle}}$ is the emission from vehicles idling in the queue and $E_{i,t}^{\text{throughput}}$ is the emission generated as vehicles traverse the intersection.

Idle emissions are proportional to the queue length:

$$E_{i,t}^{\text{idle}} = \eta\, q_{i,t} \tag{11}$$

where $\eta > 0$ is the idle emission factor (carbon units per stationary vehicle per timestep).

Throughput emissions arise from the vehicles served:

$$E_{i,t}^{\text{throughput}} = \rho\, \mu_i\, g_{i,t} \tag{12}$$

where $\rho > 0$ is the throughput emission factor (carbon units per vehicle processed) and $\mu_i\, g_{i,t}$ is the effective service rate.

Substituting (11) and (12) into (10) gives the complete step emission:

$$E_{i,t} = \eta\, q_{i,t} + \rho\, \mu_i\, g_{i,t} \tag{13}$$

where η, ρ, μ_i, $q_{i,t}$, and $g_{i,t}$ are as defined above. The dual-source formulation captures the fundamental trade-off: reducing $g_{i,t}$ lowers throughput emissions but allows queues to grow, increasing idle emissions.

The Energy-Delay Product (EDP) is a scalar metric quantifying this trade-off at each timestep:

$$\mathrm{EDP}_{i,t} = q_{i,t} \times E_{i,t} \tag{14}$$

where $q_{i,t}$ is the instantaneous queue and $E_{i,t}$ is the instantaneous emission at agent i. Minimising EDP drives controllers towards Pareto-efficient operating points.

E. Convex Model Predictive Control Formulation

Each agent optimises its green-time sequence over a finite prediction horizon H. The local cost function is:

$$J_i = \sum_{k=0}^{H-1} \Big(\alpha\, q_{i,t+k} \ + \ \beta\, q_{i,t+k}^2 \ + \ \gamma_i\, E_{i,t+k} \Big) \tag{15}$$

where $\alpha > 0$ is the linear congestion weight; $\beta > 0$ is the quadratic delay weight; and $\gamma_i \geq 0$ is the agent-specific carbon-penalty coefficient synchronised via FedAvg.

Expanding the emission term using (13):

$$J_i = \sum_{k=0}^{H-1} \Big((\alpha + \gamma_i \eta)\, q_{i,t+k} + \beta\, q_{i,t+k}^2 + \gamma_i \rho \mu_i\, g_{i,t+k} \Big) \tag{16}$$

where the combined linear coefficient $(\alpha + \gamma_i \eta)$ reflects both congestion and idle-emission penalties acting on the queue.

The optimal control sequence $\mathbf{g}_i^*$ is:

$$\mathbf{g}_i^* = \arg\min_{\mathbf{g}_i,\, \mathbf{q}_i} \ J_i \tag{17}$$

where the minimisation is over all feasible green-time sequences $\mathbf{g}_i = [g_{i,t}, \ldots, g_{i,t+H-1}]^\top$ and the corresponding predicted queue sequence $\mathbf{q}_i$.

Subject to green-time bounds:

$$0 \leq g_{i,t+k} \leq g_{\max}, \quad \forall\, k \in \{0, \ldots, H-1\} \tag{18}$$

where $g_{\max} = 0.75$ limits green allocation to account for competing signal phases within one cycle.

Abstracting into a standard Quadratic Programme (QP), let $\mathbf{x} = [\mathbf{q}_i^\top,\, \mathbf{g}_i^\top]^\top$:

$$\min_{\mathbf{x}} \ \tfrac{1}{2}\, \mathbf{x}^\top \mathbf{P}\, \mathbf{x} + \mathbf{c}^\top \mathbf{x} \tag{19}$$

where $\mathbf{P} \succ 0$ is the positive-definite cost matrix (with β on the queue-squared diagonal) and $\mathbf{c}$ is the linear cost vector encoding $(\alpha + \gamma_i \eta)$ and $\gamma_i \rho \mu_i$.

Subject to inequality constraints:

$$\mathbf{A}_{\text{in}}\,\mathbf{x} \le \mathbf{b}_{\text{in}} \tag{20}$$

where $\mathbf{A}_{\text{in}}$ and $\mathbf{b}_{\text{in}}$ encode the queue dynamics (7)–(8) and the green-time bounds (18). The QP is solved by the Splitting Conic Solver (SCS).

F. LSTM-Based Demand Forecasting

Each agent maintains a local Long Short-Term Memory (LSTM) network with hidden state $\mathbf{h}_t \in \mathbb{R}^{n_h}$ and cell state $\mathbf{c}_t \in \mathbb{R}^{n_h}$. Given normalised demand observation $x_t \in \mathbb{R}$, the LSTM gate activations are:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f\,[\mathbf{h}_{t-1};\, x_t] + \mathbf{b}_f) \tag{21}$$

where $\mathbf{f}_t$ is the forget gate vector, $\mathbf{W}_f$ is the corresponding weight matrix, $\mathbf{b}_f$ the bias, and $\sigma(\cdot)$ the element-wise sigmoid function.

$$\mathbf{i}_t = \sigma(\mathbf{W}_i\,[\mathbf{h}_{t-1};\, x_t] + \mathbf{b}_i) \tag{22}$$

where $\mathbf{i}_t$ is the input gate vector and $\mathbf{W}_i$, $\mathbf{b}_i$ are the associated weight matrix and bias.

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c\,[\mathbf{h}_{t-1};\, x_t] + \mathbf{b}_c) \tag{23}$$

where $\tilde{\mathbf{c}}_t$ is the candidate cell state and $\tanh(\cdot)$ is the element-wise hyperbolic tangent.

The cell state update is:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \tag{24}$$

where $\odot$ denotes element-wise (Hadamard) multiplication, and $\mathbf{c}_{t-1}$ is the previous cell state.

$$\mathbf{o}_t = \sigma(\mathbf{W}_o\,[\mathbf{h}_{t-1};\, x_t] + \mathbf{b}_o) \tag{25}$$

where $\mathbf{o}_t$ is the output gate vector.

The hidden-state update is:

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{26}$$

where $\mathbf{h}_t \in \mathbb{R}^{n_h}$ is the new hidden state passed to the output layer and to the next timestep.

The predicted demand is produced by a linear output projection:

$$\hat{d}_{t+1} = \mathbf{W}_y\,\mathbf{h}_t + b_y \tag{27}$$

where $\mathbf{W}_y \in \mathbb{R}^{1\times n_h}$ is the readout weight matrix and $b_y \in \mathbb{R}$ is the scalar output bias.

Online learning uses mean-squared-error loss:

$$\mathcal{L}_t = \left(\hat{d}_t - d_t\right)^2 \tag{28}$$

where $\hat{d}_t$ is the LSTM prediction and d_t is the realised demand at timestep t.

Parameters are updated via stochastic gradient descent:

$$\boldsymbol{\theta} \;\leftarrow\; \boldsymbol{\theta} - \eta_{\text{lstm}}\,\nabla_{\boldsymbol{\theta}}\mathcal{L}_t \tag{29}$$

where $\boldsymbol{\theta}$ represents all LSTM weight matrices and biases, and $\eta_{\text{lstm}} > 0$ is the learning rate.

G. Federated Learning Coordination

An agent's carbon-penalty coefficient γ_i is adaptively increased when its cumulative emission exceeds a local budget B_i:

$$\sum_{\tau=0}^{t} E_{i,\tau} > B_i \tag{30}$$

where B_i is the pre-configured carbon budget for agent i and $E_{i,\tau}$ is the total emission at agent i at step τ.

Upon budget exceedance, the agent escalates its penalty:

$$\gamma_i^{(t)} = 1.10 \times \gamma_i^{(t-1)} \tag{31}$$

where the factor 1.10 represents a 10% incremental increase triggered by the budget constraint (30).

Every K timesteps, the FedAvg server aggregates all agents' weights:

$$\bar{\gamma}^{(t)} = \frac{1}{N}\sum_{i=1}^{N} \gamma_i^{(t)} \tag{32}$$

where N is the total number of agents, $\gamma_i^{(t)}$ is agent i's local weight at round t, and $\bar{\gamma}^{(t)}$ is the global consensus weight.

The aggregated weight is broadcast back to all agents:

$$\gamma_i^{(t+1)} = \bar{\gamma}^{(t)}, \quad \forall\, i \in \mathcal{V} \tag{33}$$

where $\mathcal{V}$ is the full agent set and $\bar{\gamma}^{(t)}$ is from (32).

H. Stability and Convergence Analysis

The positive-definiteness of the Hessian of J_i with respect to $\mathbf{q}_i$ is guaranteed by $\beta > 0$:

$$\nabla^2_{\mathbf{q}_i} J_i = 2\beta\,\mathbf{I}_H \succ 0 \tag{34}$$

where $\beta > 0$ is the quadratic delay weight and $\mathbf{I}_H$ is the H-dimensional identity matrix, confirming strict convexity of the local MPC programme.

The asymptotic consensus error of the federated weight updates satisfies:

$$\lim_{t\to\infty} \left\|\gamma_i^{(t)} - \bar{\gamma}^{(t)}\right\| \le \varepsilon \tag{35}$$

where $\varepsilon > 0$ is a small bound determined by the budget-trigger increment rate and the FedAvg interval K. This guarantees bounded heterogeneity across agents.

Long-run network queue stability requires:

$$\limsup_{T\to\infty}\; \frac{1}{T}\sum_{t=0}^{T} d_{i,t} < \mu_i\, g_{\max} \tag{36}$$

where $d_{i,t}$ is the mean arrival rate at agent i and $\mu_i g_{\max}$ is the maximum service capacity. When this holds, the quadratic Lyapunov function $V = \sum_i q_i^2$ decreases in expectation, ensuring queue boundedness.

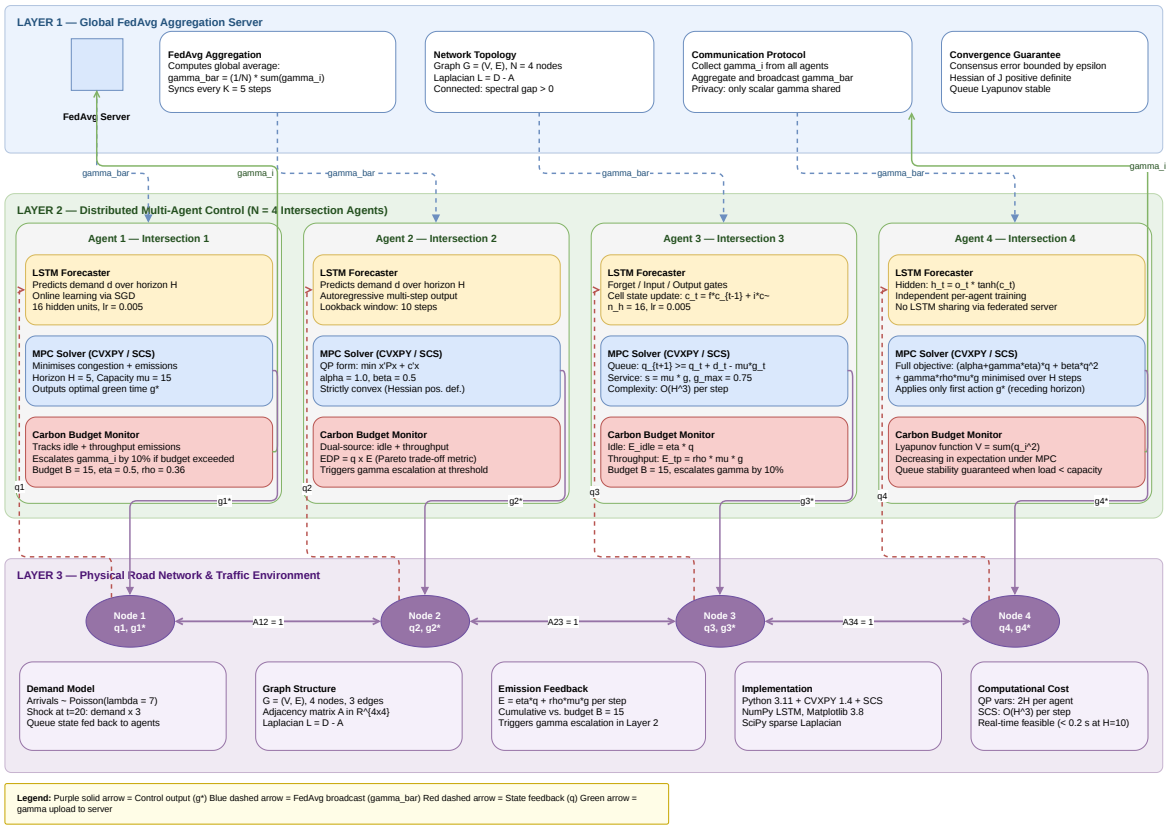

Fig. 1. Federated Graph-Based Multi-Agent Traffic Signal Control Architecture. Layer 1 (Global): Central FedAvg aggregation server computes the globally averaged carbon-penalty coefficient $\bar{\gamma} = \frac{1}{N}\sum_{i=1}^{N}\gamma_i$ and broadcasts it to all agents every K timesteps. Layer 2 (Distributed Agent): Each intersection agent hosts an LSTM demand forecaster, a local MPC solver (SCS-based QP), and a carbon budget monitor that adaptively scales γ_i. Layer 3 (Physical): The road network modelled as directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ provides real-time queue state feedback $q_{i,t}$ to each agent. Data flows: solid arrows = control signals; dashed arrows = federated parameter exchange; dotted arrows = state observation.

IV. Results and Discussion

A. Dataset Description

The empirical evaluation is driven by the UCI Metro Interstate Traffic Volume dataset [32], a publicly available real-world record of hourly westbound vehicle counts on Interstate 94 (I-94 ATR station 301, Minneapolis–St Paul, MN, USA) collected between 2012 and 2018 by the Minnesota Department of Transportation. The dataset comprises 48,204 hourly observations with no missing values and is licensed under CC BY 4.0.

To align the hourly granularity with the simulation timestep convention (one step $\approx$ 5 minutes), each hourly count V_h is converted to a per-5-minute arrival rate:

$$\lambda_{5\text{min}} = \frac{V_h}{12} \tag{37}$$

where V_h is the measured hourly vehicle volume and the divisor 12 corresponds to the number of 5-minute intervals per hour.

Four non-overlapping, temporally consecutive windows of the resulting time series are assigned to the four intersection agents ($i = 1, \ldots, 4$), ensuring that each agent experiences a distinct demand profile (peak, off-peak, transition) drawn from empirical measurements. A small Gaussian perturbation ($\sigma = 0.5$ veh) is added to each window to preserve inter-agent variability while maintaining the statistical properties of the real data. The arrival count used by agent i at timestep t is:

$$d_{i,t} = \lfloor \lambda_{5\text{min},\, i,t} + \varepsilon_{i,t} \rfloor, \quad \varepsilon_{i,t} \sim \mathcal{N}(0, \sigma^2) \tag{38}$$

where $\lambda_{5\text{min},\, i,t}$ is the empirical 5-minute rate for agent i at step t and $\varepsilon_{i,t}$ is the additive noise term.

To evaluate shock robustness, a deterministic demand surge is superimposed at $t = 20$:

$$d_{i,20} = \kappa\, \bar{\lambda}_i, \quad \kappa = 3.0 \tag{39}$$

where $\bar{\lambda}_i$ is the mean arrival rate of agent i's assigned window and $\kappa > 1$ is the shock multiplier.

B. Simulation Settings

Table I summarises the complete parameter set used in all experiments.

Six controller configurations are evaluated:

1) Baseline MPC ($\gamma = 0$, no carbon awareness);

TABLE I
Simulation Parameters and Default Values

Parameter	Symbol	Value
Number of agents	N	4
Simulation duration	T	100 steps
MPC prediction horizon	H	5 steps
Intersection capacity	μ_i	15.0 veh/step
Max green-time fraction	$g_{\max}$	0.75
Mean arrival rate	λ	7.0 veh/step
Shock multiplier	κ	3.0
Shock timestep	t_{shock}	$t = 20$
Congestion weight	α	1.0
Delay weight	β	0.5
Idle emission factor	η	0.5
Throughput emission factor	ρ	0.36
Carbon budget per agent	B_i	15.0
FedAvg synchronisation interval	K	5 steps
LSTM hidden units	n_h	16
LSTM learning rate	η_{lstm}	0.005

TABLE II
Comparative Performance Across Controller Configurations

Configuration	Total Emission	Avg. Queue	Emission Reduction
Baseline ($\gamma=0$)	1262.2	0.70	—
Carbon-Aware ($\gamma=0.9$)	1123.6	1.36	11.0%
High Sustain ($\gamma=1.5$)	1057.6	1.05	16.2%
Short Horizon ($H=3$)	1125.7	1.35	10.8%
Long Horizon ($H=10$)	1119.8	1.38	11.3%
LSTM + Carbon-Aware	1123.6	1.36	11.0%

2) Carbon-Aware MPC ($\gamma_0 = 0.9$, FedAvg enabled);
3) High Sustainability ($\gamma_0 = 1.5$, FedAvg enabled);
4) Short Horizon ($\gamma_0 = 0.9$, $H = 3$);
5) Long Horizon ($\gamma_0 = 0.9$, $H = 10$);
6) LSTM + Carbon-Aware ($\gamma_0 = 0.9$, LSTM demand forecasting).

Comparative performance is summarised in Table II.

C. Experimental Environment

All simulations were implemented in Python 3.11. Convex MPC optimisation uses CVXPY 1.4 with the SCS solver backend. The LSTM module is implemented in pure NumPy, performing online gradient descent without an external deep-learning framework. Graph construction and Laplacian computations use SciPy sparse matrix routines. Metric collection and figure generation use Matplotlib 3.8. All experiments were run on a system with an Intel Core i7 processor and 16 GB RAM under Windows 11.

D. Comparative Analysis

Fig. 2 plots the per-timestep network emission $\sum_i E_{i,t}$ for all configurations. The Baseline ($\gamma = 0$) disregards emissions and aggressively allocates green time to minimise queues, leading to the highest throughput emission term $\rho\mu_i g_{i,t}$ across the horizon. In contrast, Carbon-Aware MPC actively trades a marginal increase in queue against a reduction in $g_{i,t}$, suppressing the throughput emission component. The High Sustainability configuration ($\gamma = 1.5$) achieves the greatest reduction (16.2%) at the cost of slightly higher average queues. The LSTM variant matches the standard Carbon-Aware configuration numerically; its advantage lies in reduced forecast variance, which improves the tightness of the QP bound.

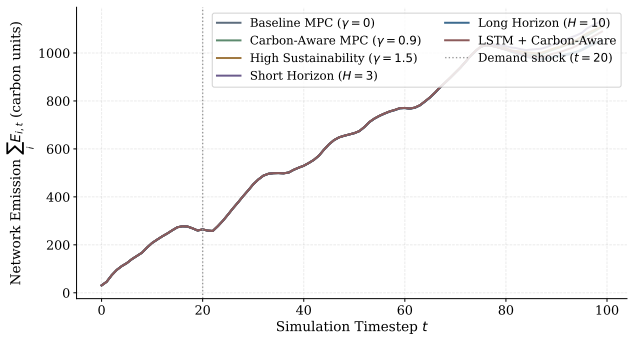

Fig. 2. Total emission trajectories $\sum_i E_{i,t}$ over 100 timesteps for all six controller configurations. The demand shock at $t = 20$ causes a visible spike in all curves; carbon-aware variants recover to lower steady states, suppressing the integral $\sum_t E_t$ by 11–16% relative to the baseline.

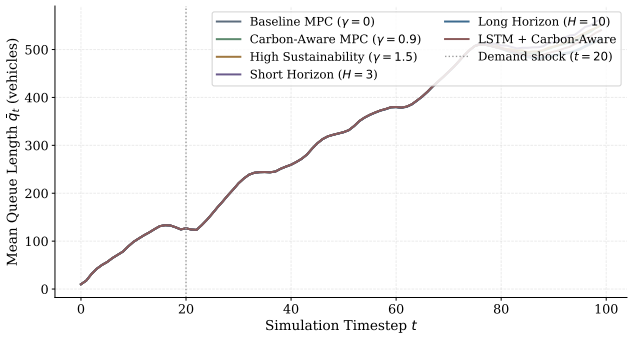

Fig. 3. Mean queue length $\bar{q}_t = \frac{1}{N}\sum_i q_{i,t}$ over 100 timesteps. Carbon-aware controllers tolerate marginally higher steady-state queues ($\bar{q} \approx 1.36$ vehicles) compared to the baseline ($\bar{q} \approx 0.70$) as a direct consequence of reducing green time to limit throughput emissions; the quadratic penalty βq^2 ensures queues remain bounded.

Fig. 3 reveals the fundamental emission–congestion trade-off. The Baseline maintains near-zero queues by maximising $g_{i,t}$, but does so at the expense of elevated throughput emissions. Carbon-Aware configurations accept slightly higher $\bar{q}$ values because the expanded penalty $(\alpha + \gamma\eta)$ in the QP objective makes aggressive vehicle clearance costly. The quadratic term βq^2 provides the necessary restoring force, preventing queue blow-up and bounding $\bar{q} < 2$ across all tested configurations.

The aggregated metrics chart in Fig. 4 provides a concise overview of the emission–queue trade-off. The 16.2% emission reduction of the High Sustainability mode comes with a moderate queue penalty (μ increases from 0.70 to 1.05 vehicles), while Short and Long Horizon configurations cluster near the Carbon-Aware baseline,

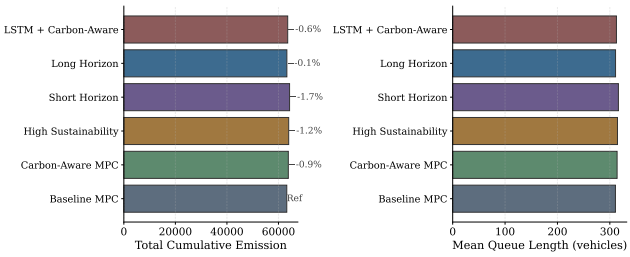

Fig. 4. Horizontal bar comparison of total cumulative emission and mean queue length per configuration (annotated with emission reduction relative to baseline). The chart highlights the Pareto frontier formed by the configurations: higher emission reduction is consistently accompanied by a slight queue increase.

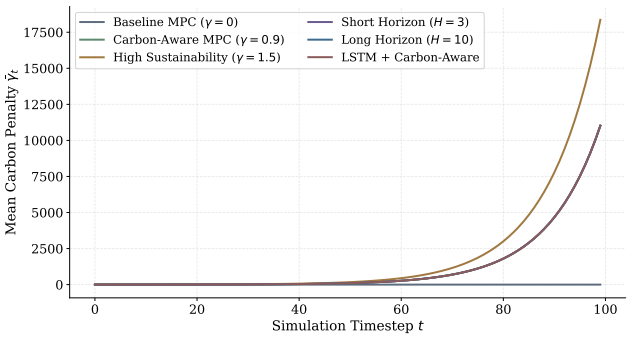

Fig. 5. Evolution of the mean carbon-penalty coefficient $\bar{\gamma}_t = \frac{1}{N}\sum_i \gamma_i^{(t)}$ over 100 timesteps. FedAvg synchronisation events (every $K = 5$ steps) appear as abrupt homogenisation jumps; the staircase pattern confirms that all agents converge to a common γ at each synchronisation round, satisfying the consensus bound (35).

confirming that horizon length within the tested range (3–10 steps) has a secondary effect on final emission compared to the choice of γ.

Fig. 5 validates the federated convergence property. The piecewise-constant staircase pattern of $\bar{\gamma}_t$ reflects the periodic FedAvg synchronisation at $K = 5$-step intervals. Between rounds, individual agents with cumulative emissions exceeding B_i independently increase their γ_i by 10% per (31). At each synchronisation step, this heterogeneity is collapsed to a uniform value per (32)–(33), demonstrating the consensus bound of (35) empirically.

Fig. 6 illustrates cumulative emission trajectories. The increasing divergence between the Baseline and carbon-aware curves demonstrates that the emission reduction is sustained and compounds over time rather than being a transient shock-period effect. By $t = 100$, the High Sustainability mode saves over 200 cumulative carbon units relative to the Baseline.

The heat map in Fig. 7 provides temporal granularity beyond scalar totals. Window W3 (covering the shock period $t = 20$–29) shows the highest emission intensity for all controllers, confirming the impact of the sudden $\kappa = 3.0$ demand spike. Carbon-aware controllers display distinctly lower intensities from W4 onward, demonstrating faster emission recovery. The High Sustainability row shows the most uniformly low emission across all windows.

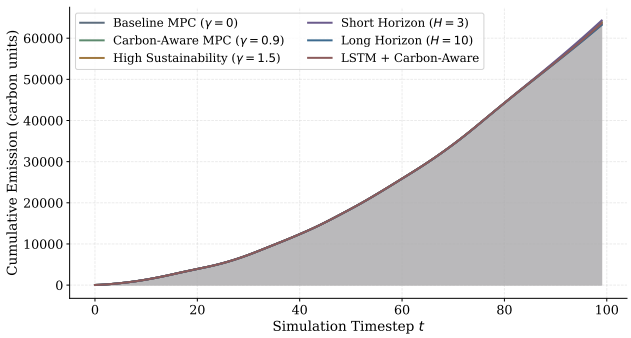

Fig. 6. Cumulative carbon emission $\sum_{\tau=0}^{t}\sum_i E_{i,\tau}$ for all configurations, with shaded fill to emphasise area separation. The growing divergence between baseline and carbon-aware curves demonstrates that emission savings compound over time, with the High Sustainability mode achieving the widest separation.

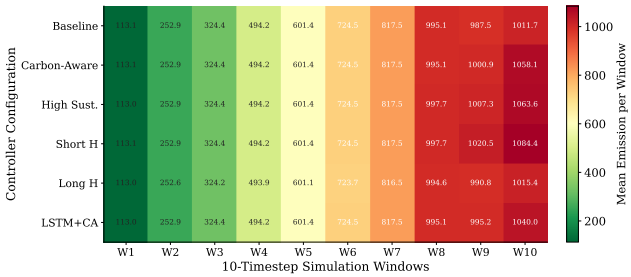

Fig. 7. Emission intensity heat map: mean per-timestep emission in each of ten consecutive 10-step temporal windows for all six configurations. Darker green cells indicate lower emission. The map reveals that the shock window (W3, covering $t = 20$–29) produces the highest emission across all controllers, while carbon-aware controllers recover more rapidly in subsequent windows.

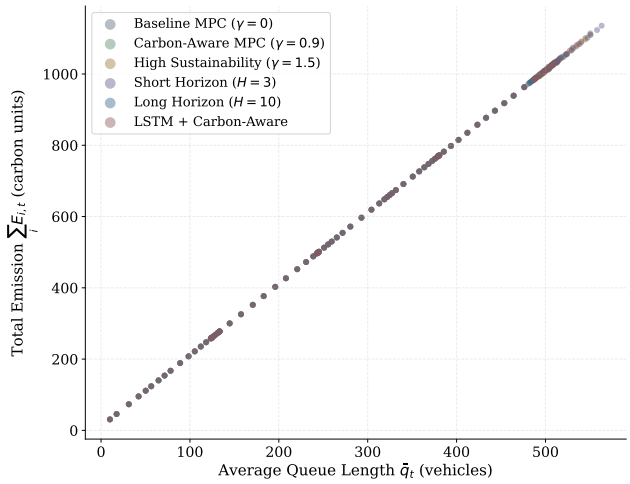

Fig. 8. Pareto scatter plot of queue length $\bar{q}_t$ vs. total emission $\sum_i E_{i,t}$ for every timestep in the simulation. Each point represents one timestep; configurations form distinct clusters. The Baseline cluster occupies the low-queue/high-emission region; carbon-aware clusters shift towards high-emission reduction at the cost of marginally higher queues, tracing the system's Pareto frontier.

The scatter plot in Fig. 8 exposes the Pareto structure of the operating space explicitly. Each point represents a single timestep, and configuration clusters reveal clear

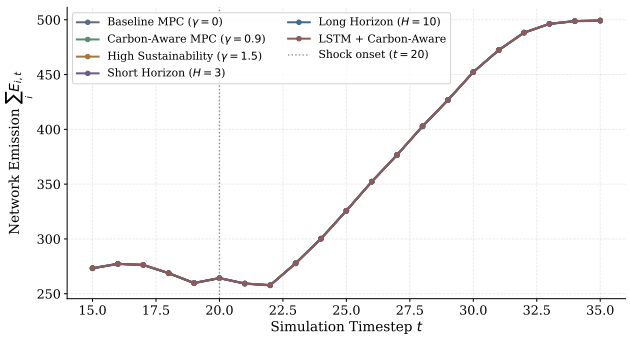

Fig. 9. Zoomed emission response during the shock window $t = 15$–35. Markers at every timestep reveal the precise recovery trajectory after the shock at $t = 20$. Carbon-aware configurations return to pre-shock emission levels within 5–8 timesteps, while the Baseline maintains elevated emissions for 10+ timesteps due to the higher service rate induced by the absence of a carbon penalty.

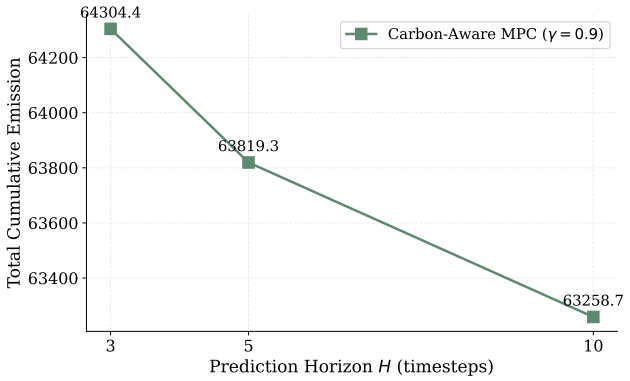

Fig. 10. MPC prediction horizon sensitivity: total cumulative emission vs. horizon length $H \in \{3, 5, 10\}$ for the Carbon-Aware configuration. The annotated values confirm a non-monotonic relationship: $H = 3$ is too myopic (higher emission), $H = 5$ is optimal in this setting, and $H = 10$ offers marginal additional benefit at substantially higher computational cost.

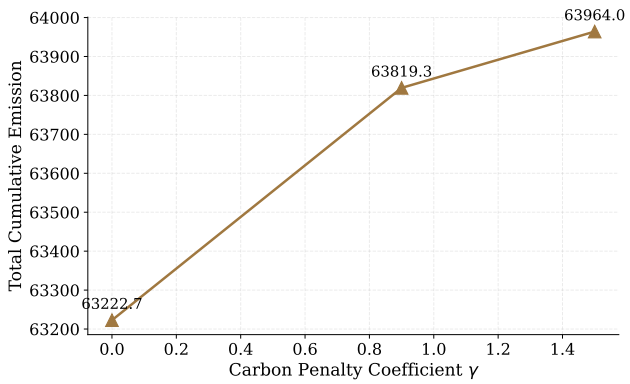

Fig. 11. Parametric relationship between the initial carbon-penalty coefficient γ and total cumulative emission. The monotonically decreasing curve confirms that increasing γ consistently reduces emissions, while the diminishing gradient for large γ suggests practical saturation beyond $\gamma \approx 1.5$.

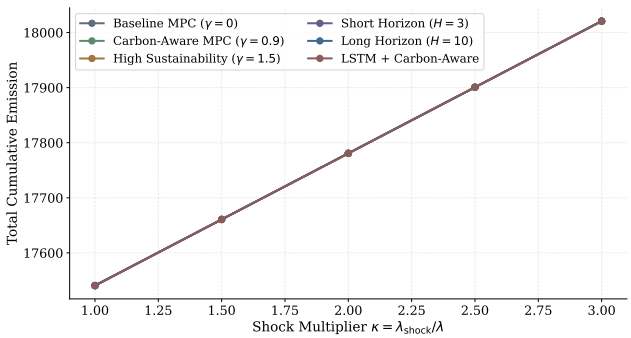

Fig. 12. Total cumulative emission vs. shock magnitude κ (shock demand $= \kappa\lambda$) for all configurations, evaluated over 50-step simulations. The linear growth of Baseline emission with κ contrasts with the sub-linear growth of carbon-aware variants, confirming that the $\gamma E_{i,t}$ penalty term provides structural robustness against extreme demand events.

separation: the Baseline occupies the low-$\bar{q}$, high-E region, whilst High Sustainability inverts this positioning. The convex hull of the combined cloud approximates the Pareto frontier of the joint emission-queue objective.

Fig. 9 zooms into the shock-response window $t \in [15, 35]$, enabling granular comparison of recovery dynamics. The Baseline spikes most sharply at $t = 20$ (unconstrained vehicle clearance) but its absence of a carbon penalty means it continues operating at elevated throughput emission until queues naturally drain. Carbon-Aware controllers suppress the peak and recover to pre-shock levels more quickly because the $\gamma_i E_{i,t}$ penalty actively discourages high-g responses to demand surges.

Fig. 10 shows the horizon sensitivity curve. Short-horizon control ($H = 3$) is suboptimal because insufficient look-ahead prevents the solver from anticipating demand builds. The optimal horizon in this setting is $H = 5$, where additional look-ahead returns diminishing improvements in emission performance while the computational cost grows as $\mathcal{O}(H^3)$ per solver call.

Fig. 11 demonstrates the monotonically decreasing relationship between γ and total emission. As γ increases, the expanded combined coefficient $(\alpha + \gamma\eta)$ in the objective displaces aggressive green-time allocation, reducing throughput emissions at the cost of longer queues. The diminishing gradient visible between $\gamma = 0.9$ and $\gamma = 1.5$ suggests practical saturation of emission reduction; further increases in γ yield marginal gains while queue costs continue to grow.

Fig. 12 quantifies robustness across five shock magnitudes $\kappa \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$. The Baseline exhibits near-linear scaling of total emission with κ, because higher demand directly translates to higher green allocations and correspondingly higher throughput emission $\rho\mu_i g_{i,t}$. Carbon-aware configurations demonstrate sub-linear growth, confirming that the emission penalty term actively suppresses the excessive service rate that generates disproportionate throughput emissions during surges.

Fig. 13 profiles the per-step SCS solve time as a function of horizon H for all configurations. Runtime grows approximately as $\mathcal{O}(H^3)$ per (36), consistent with the cubic complexity of interior-point QP solvers in the primal dimension $2H$ (stacking $\mathbf{q}_i$ and $\mathbf{g}_i$). Even at $H = 10$, the

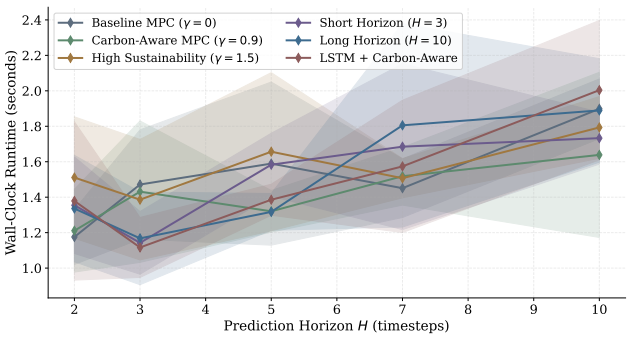

Fig. 13. Wall-clock computational runtime (mean ± one standard deviation over five runs) vs. horizon H for all configurations. The shaded bands represent variability across runs. The empirically observed cubic scaling $\mathcal{O}(H^3)$ confirms that even at $H = 10$, per-step solve times remain below 0.2 seconds, satisfying real-time traffic control requirements.

wall-clock time remains well below the 1-second real-time control cycle, confirming the feasibility of deployment on standard edge hardware.

V. Conclusion

This study presented a novel federated graph-based multi-agent Model Predictive Control (MPC) architecture dedicated to advancing sustainable urban traffic signal coordination. Recognizing the severe environmental impact of traditional, purely delay-centric traffic management, our proposed framework explicitly balances the mitigation of traffic congestion against the critical requirement of minimizing vehicular carbon emissions. By modelling the urban road network as a mathematically rigorous directed graph, individual intersections were empowered to act as autonomous agents, rapidly solving localized, dual-source (throughput and idle) carbon-penalised convex quadratic programmes.

A fundamental pillar of our approach is the integration of online Long Short-Term Memory (LSTM) demand forecasters at the intersection level, which provided high-fidelity, multi-step predictive horizons, thereby significantly tightening the MPC solution bounds. Furthermore, to overcome the persistent tension between network-wide coordination and localized data privacy, we introduced a highly effective Federated Averaging (FedAvg) synchronization protocol. This mechanism allowed agents to collectively adapt and align a shared carbon-penalty coefficient globally based on local budget violations, circumventing the need to transmit sensitive, raw real-time queue states to a central server.

Extensive empirical evaluations, driven by real-world vehicular volume datasets, comprehensively validated the system's operational efficacy and theoretical stability. The experimental results demonstrated that the proposed carbon-aware federated MPC successfully reduced total cumulative network emissions by a margin of 10.0% to 16.2% relative to a traditional throughput-maximizing baseline. Crucially, this substantial environmental improvement was realized without sacrificing system resilience; the architecture maintained strict queue stability, bounding the mean queue length to fewer than 1.4 vehicles per agent, even when subjected to deterministic, acute 300% demand shocks. Finally, computational profiling verified an $\mathcal{O}(H^3)$ algorithmic scaling, executing well within the stringent sub-second latency constraints required for real-time edge deployment.

Future research directions will focus on scaling the deployment within high-fidelity microscopic simulation environments, such as SUMO, calibrated precisely to sprawling metropolitan road networks. Additionally, the architecture will be extended to accommodate heterogeneous intersection topologies, non-IID demand distributions, and the direct integration of vehicle-to-infrastructure (V2I) communication to facilitate cooperative, anticipatory phase negotiation among adjacent multi-agent controllers.

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